

离散回归II

Ordered Probit Model

In some cases, the variable to be modeled has a natural *ordinal* interpretation.

Some examples include:

- ① Education, measured categorically, (e.g. 1 = < HS, 2 = HS, 3 = Some college, etc.).
 - ② Income, also measured categorically.
 - ③ Survey responses, coded as a degree of opinion (e.g. 1 = Strongly Disagree, 2 = Disagree, 3 = Agree, 4 = Strongly Agree).
 - ④ These are in contrast to other choices such as type of insurance or selected mode of transportation, for example, that are not ordered.
- In this lecture we discuss *ordinal choice* models, and focus on the *ordered probit* in particular.

Ordered Probit Model

- Ordered probability models are derived by defining an unobserved variable z that is used as a basis for modeling the ordinal ranking of data.

$$z = \beta X + \varepsilon$$

where X is a vector of variables determining the discrete ordering for observation n , β is a vector of estimable parameters, and ε is a random disturbance.

Ordered Probit Model

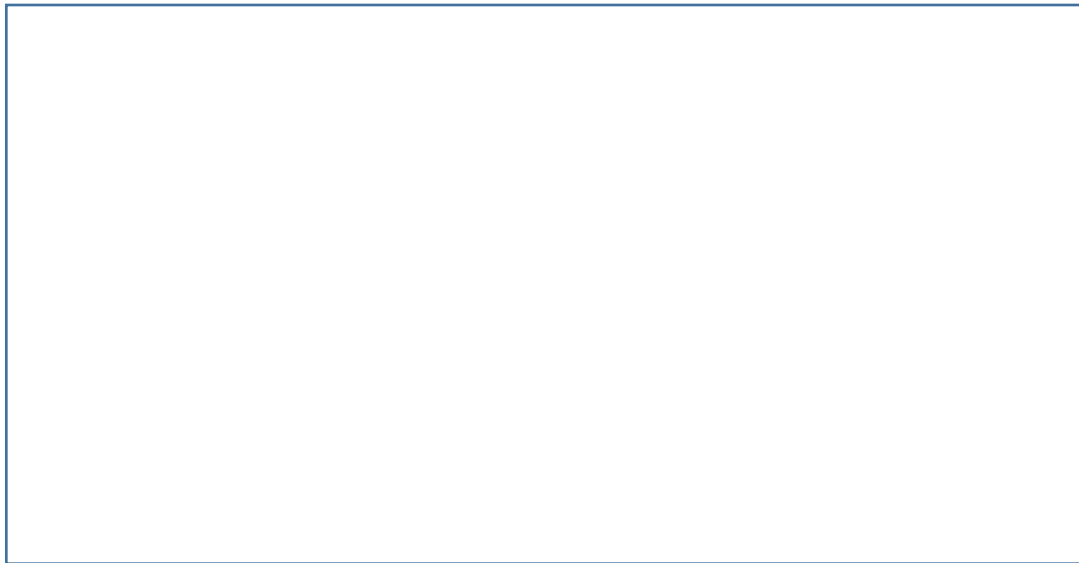
Using this equation, observed ordinal data, y , for each observation are defined as

$$\begin{aligned} y &= 1 && \text{if } z \leq \mu_0 \\ y &= 2 && \text{if } \mu_0 < z \leq \mu_1 \\ y &= 3 && \text{if } \mu_1 < z \leq \mu_2 \\ y &= \dots \\ y &= I && \text{if } z \geq \mu_{I-1} \end{aligned}$$

where μ are estimable parameters (referred to as thresholds) that define y , which corresponds to integer ordering, and I is the highest integer ordered response.

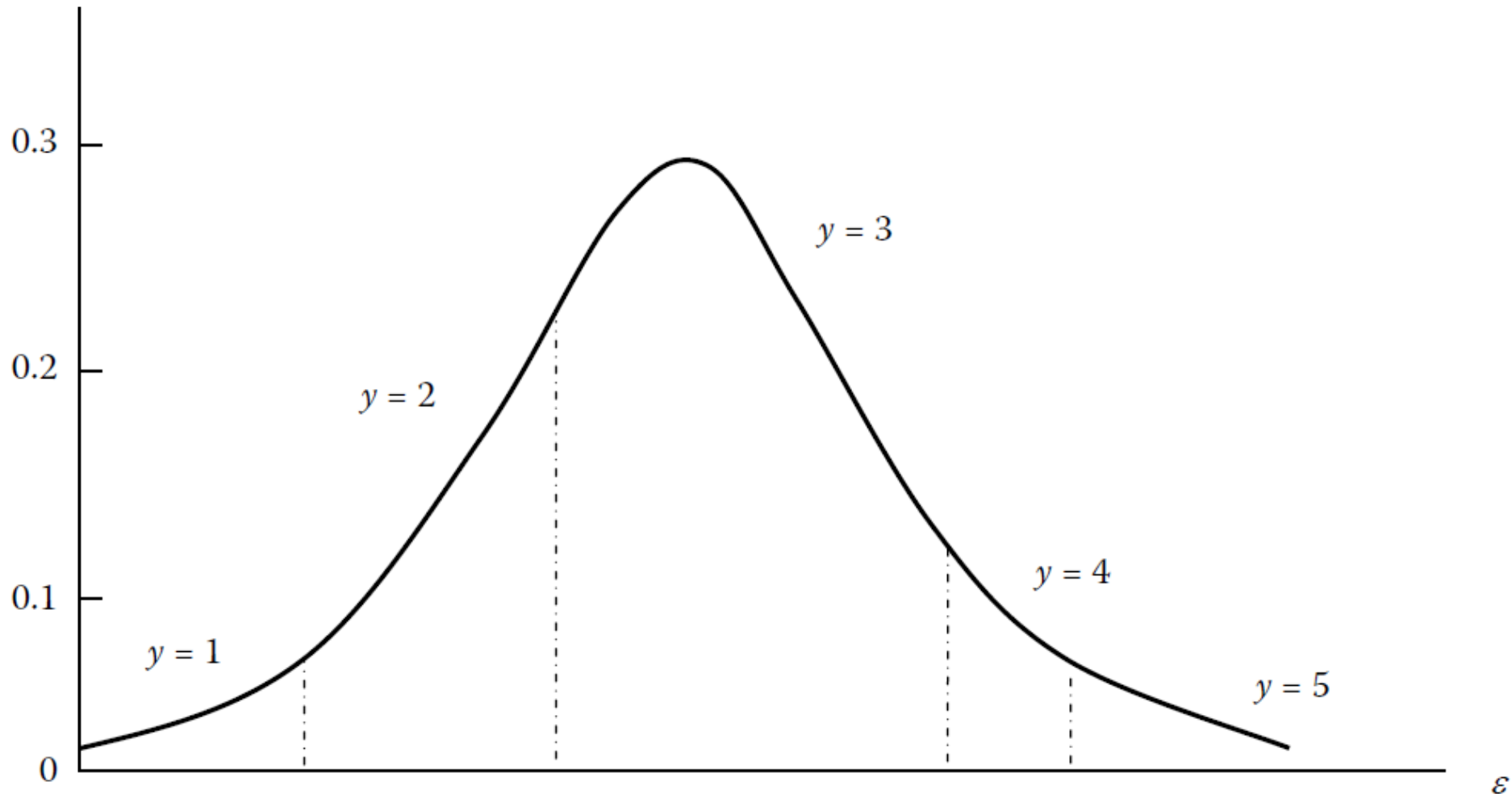
Ordered Probit Model

- ❑ The μ are parameters that are estimated jointly with the model parameters.
- ❑ If ε are assumed to be normally distributed across observations with mean = 0 and variance = 1, an ordered probit model results with the ordered selection probabilities being



Ordered Probit Model

$f(\varepsilon)$



Ordered Probit Model

- Threshold μ_0 is set equal to 0.
- For estimation, the above is written as

$$P(y = i) = \Phi(\mu_i - \beta X) - \Phi(\mu_{i+1} - \beta X)$$

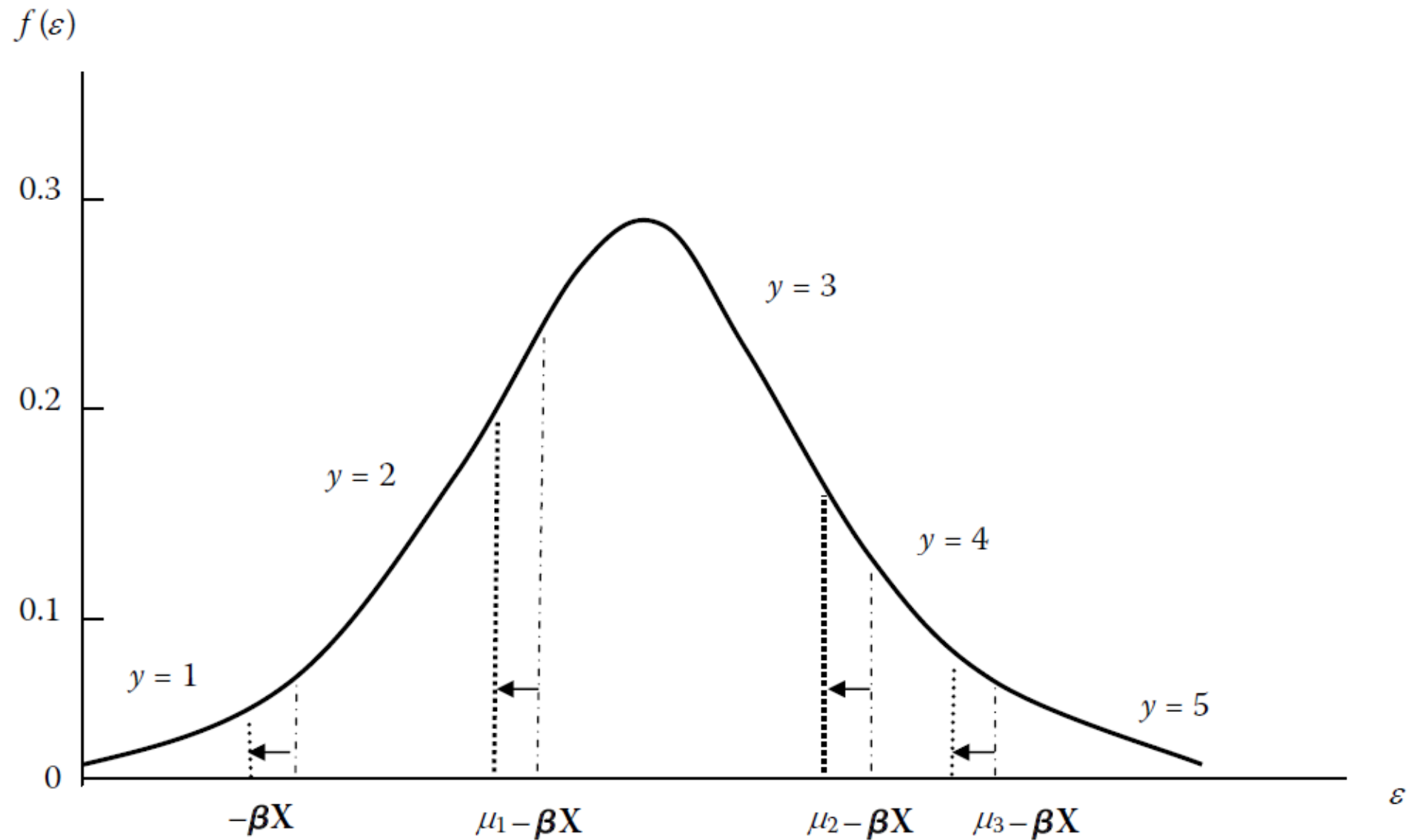
- The likelihood function is (over the population of N observations)

$$L(y | \beta_1, \dots, \beta_k, \mu_2, \dots, \mu_{I-1}) = \prod_{n=1}^N \prod_{i=1}^I [\Phi(\mu_i - \beta X_n) - \Phi(\mu_{i+1} - \beta X_n)]^{\delta_{in}}$$

$$LL = \sum_{n=1}^N \sum_{i=1}^I \delta_{in} \ln [\Phi(\mu_i - \beta X_n) - \Phi(\mu_{i+1} - \beta X_n)]$$

where δ_{in} is equal to 1 if the observed discrete outcome for observation n is i , and zero otherwise.

Ordered Probit Model



Ordered Probit Model

- To obtain a sense of the direction of the effects on the interior categories, marginal effects are computed for each category

$$\frac{\partial P(y = 1)}{\partial \mathbf{X}} = -\phi(-\boldsymbol{\beta}\mathbf{X})\boldsymbol{\beta}'$$

$$\frac{\partial P(y = 2)}{\partial \mathbf{X}} = [\phi(\mu_0 - \boldsymbol{\beta}\mathbf{X}) - \phi(\mu_1 - \boldsymbol{\beta}\mathbf{X})]\boldsymbol{\beta}'$$

$$\frac{\partial P(y = 3)}{\partial \mathbf{X}} = [\phi(\mu_1 - \boldsymbol{\beta}\mathbf{X}) - \phi(\mu_2 - \boldsymbol{\beta}\mathbf{X})]\boldsymbol{\beta}'$$

$$\frac{\partial P(y = \dots)}{\partial \mathbf{X}} = \dots$$

$$\frac{\partial P(y = I)}{\partial \mathbf{X}} = -\phi(\mu_{I-2} - \boldsymbol{\beta}\mathbf{X})\boldsymbol{\beta}'$$

Ordered Probit Estimation Results for the Question “High-Occupancy Vehicle (HOV) Lanes Should be Open to All Vehicles, Regardless of Vehicle Occupancy Level” (Disagree Strongly, Disagree, Neutral, Agree, Agree Strongly)

Independent Variable	Estimated Parameter	<i>t</i> -Statistic
Constant	0.707	−3.92
Drive alone indicator (1 if commuter’s usual mode is drive alone, 0 otherwise)	1.144	7.02
Flexible work-time indicator (1 if commuter has flexible work hours, 0 other wise)	0.283	2.06
High-income indicator (1 if commuter’s household income is greater than \$100,000 per year, 0 otherwise)	0.365	1.59
Older age indicator (1 if commuter is 50 years old or older, 0 otherwise)	0.249	1.43
Number of times in the past 5 commutes the commuter changed from normal routes or departure time	0.081	1.59
Threshold 1	0.644	8.61
Threshold 2	0.890	10.55
Threshold 3	1.27	13.10
Number of observations	281	
Log-likelihood at zero	−424.05	
Log-likelihood at convergence	−395.65	

Variable	Marginal Effects				
	[$y = 1$]	[$y = 2$]	[$y = 3$]	[$y = 4$]	[$y = 5$]
Drive alone indicator (1 if commuter's usual mode is drive alone, 0 otherwise)	-0.4050	-0.0456	0.0299	0.0841	0.3366
Flexible work-time indicator (1 if commuter has flexible work hours, 0 otherwise)	-0.1002	-0.0113	0.0074	0.0208	0.0833
High-income indicator (1 if commuter's household income is greater than \$100,000 per year, 0 otherwise)	-0.1291	-0.0145	0.0095	0.0268	0.1073
Older age indicator (1 if commuter is 50 years old or older, 0 otherwise)	-0.0883	-0.0099	0.0065	0.0183	0.0734
Number of times in the past 5 commutes the commuter changed from normal routes or departure time	-0.0285	-0.0032	0.0021	0.0059	0.0237